

Run report

2026-09-11T160419Z

Configuration

Parameter	Value	Parameter	Value
batch_size	1,024	n_samples	1,600,000
dim	12	dataset	jets
seed	1,851,207,501	data_seed	42
hidden_units	128	n_layers	3
n_epochs	100	n_disc_steps	5
lr_g	3.000×10^{-5}	lr_d	1.000×10^{-4}
lambda_dispersion	0.01500	mmd_subsample	16,384
mmd_test	2.573×10^{-4}	best_epoch	91
compile_cache_warm	True		
variables	m [GeV], $\ln \rho$, $\lambda_{0.5}^1$, w , λ_2^1 , z_g , $\tau_{21}^{(\beta=1)}$, M , n_{ch} , f_{ch} , p_T^D , q		
mmd_sigmas_detector	$3.107 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		
mmd_sigmas_particle	$3.027 \times (\frac{1}{2}, \frac{1}{\sqrt{2}}, 1, \sqrt{2}, 2)$		

Timing

Phase	Seconds	Share (%)	Detail
train	40.643	57.9	train
transfer	5.714		train
compile	3.910		train
epochs	29.729		train
select	1.080		train
particle_mmd	0.403	0.6	train
save	0.352	0.5	train
data	16.574	23.6	cache hit – load
load	1.756	2.5	load
plots	10.416	14.8	load
evaluate	0.008	0.0	load
total	70.152	100.0	

Detector-level metrics

Distance to Data. Improvements are relative to Simulation.

Wasserstein [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	208.9	138.8	33.6	9.648	95.4	9.340	95.5
$\ln \rho$	227.9	20.71	90.9	23.16	89.8	11.60	94.9
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	262.6	12.84	95.1	18.66	92.9	7.440	97.2
w	265.6	9.978	96.2	16.75	93.7	6.082	97.7
λ_2^1	255.4	9.215	96.4	15.26	94.0	6.440	97.5
Splitting and 2-prong substructure							
$z_g^\dagger$	60.45	60.45	0.0	15.93	73.6	17.11	71.7
$\tau_{21}^{(\beta=1)}$	121.8	70.10	42.4	37.71	69.0	18.78	84.6
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	224.2	177.3	20.9	12.98	94.2	20.78	90.7
n_{ch}	220.8	172.3	22.0	13.59	93.8	22.69	89.7
f_{ch}	52.86	40.94	22.6	5.765	89.1	7.496	85.8
p_T^D	256.8	30.80	88.0	14.45	94.4	16.82	93.4
q	79.67	17.51	78.0	3.901	95.1	5.595	93.0

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

JS divergence [$\times 10^3$]

Observable	Sim.	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	6.762	3.666	45.8	0.07863	98.8	0.06873	99.0
$\ln \rho$	8.383	0.4017	95.2	0.2312	97.2	0.1500	98.2
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	9.359	0.4714	95.0	0.1792	98.1	0.1141	98.8
w	9.139	0.4527	95.0	0.1621	98.2	0.1155	98.7
λ_2^1	7.776	0.1728	97.8	0.1488	98.1	0.1204	98.5
Splitting and 2-prong substructure							
$z_g^\dagger$	0.6723	0.6723	0.0	0.1112	83.5	0.1136	83.1
$\tau_{21}^{(\beta=1)}$	3.372	1.934	42.7	0.3410	89.9	0.1702	95.0
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	9.640	7.633	20.8	0.1401	98.5	0.1704	98.2
n_{ch}	7.512	6.314	16.0	0.1049	98.6	0.1431	98.1
f_{ch}	1.033	0.7960	22.9	0.08459	91.8	0.09115	91.2
p_T^D	8.223	1.311	84.1	0.1142	98.6	0.1381	98.3
q	1.576	0.2512	84.1	0.07234	95.4	0.08142	94.8

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

VLC divergence [$\times 10^3$]

[illegible]

Particle-level metrics

Distance to Truth. Improvements are relative to Generation.

Wasserstein [$\times 10^3$]

Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	111.7	94.07	15.8	89.36	20.0	86.32	22.8
$\ln \rho$	253.1	32.57	87.1	13.26	94.8	17.29	93.2
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	303.0	18.71	93.8	16.66	94.5	6.176	98.0
w	284.1	12.29	95.7	12.51	95.6	3.450	98.8
λ_2^1	253.8	11.13	95.6	9.086	96.4	3.172	98.8
Splitting and 2-prong substructure							
$z_g^\dagger$	113.4	113.4	0.0	25.05	77.9	24.69	78.2
$\tau_{21}^{(\beta=1)}$	259.2	150.5	41.9	45.47	82.5	11.62	95.5
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	256.8	204.8	20.2	30.94	88.0	24.19	90.6
n_{ch}	234.6	188.6	19.6	32.00	86.4	24.70	89.5
f_{ch}	45.78	45.33	1.0	4.212	90.8	5.079	88.9
p_T^D	236.3	55.36	76.6	16.46	93.0	17.61	92.5
q	79.37	18.15	77.1	3.499	95.6	5.307	93.3

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

JS divergence [$\times 10^3$]

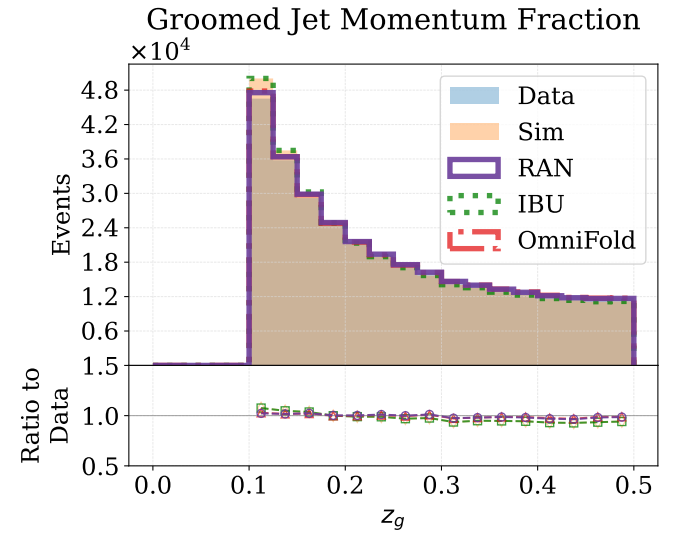
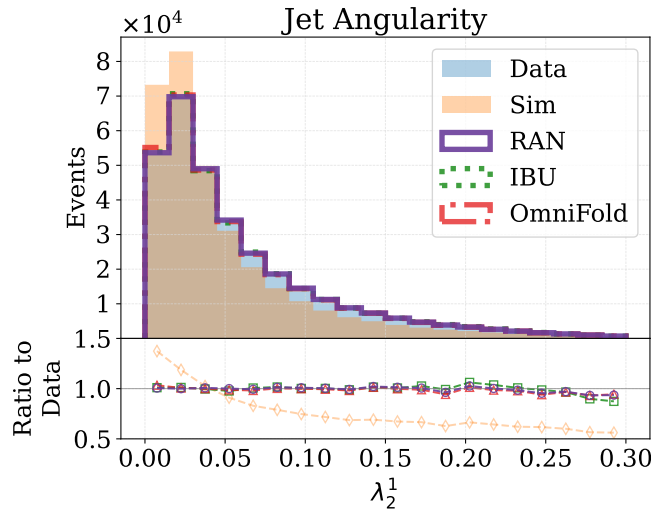
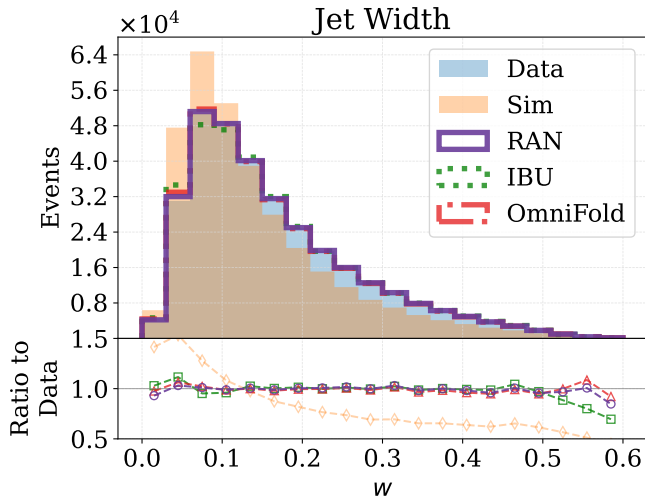
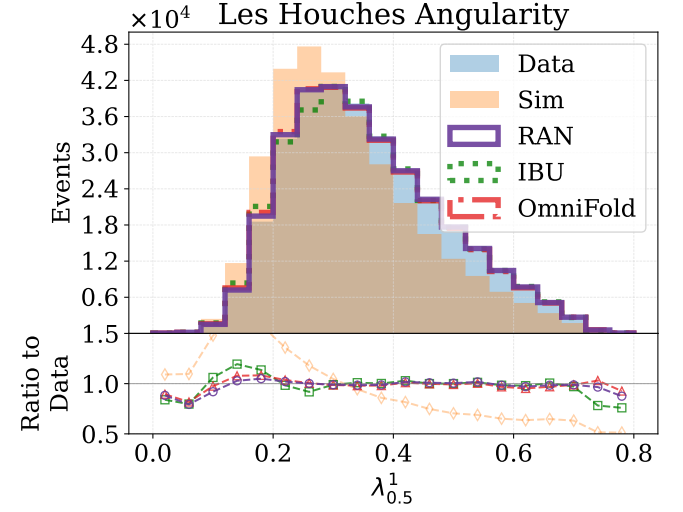
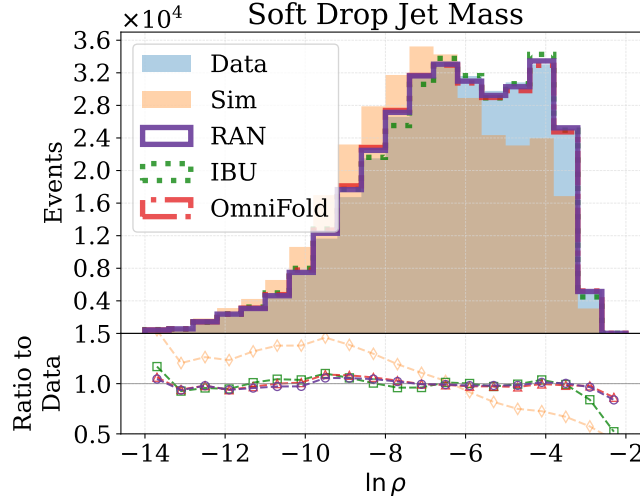
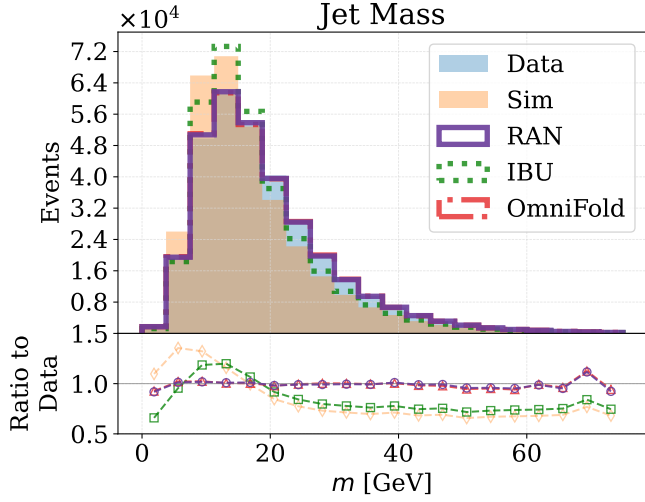
Observable	Gen	IBU	impr. (%)	OmniFold	impr. (%)	RAN	impr. (%)
Mass and hard scale (IRC-safe kinematics)							
m [GeV]	1.963	11.86	−504.3	1.650	16.0	1.596	18.7
$\ln \rho$	9.878	1.731	82.5	0.2391	97.6	0.1995	98.0
Continuous angularities (IRC-safe jet shapes)							
$\lambda_{0.5}^1$	12.34	2.009	83.7	0.2387	98.1	0.1552	98.7
w	11.39	1.465	87.1	0.2232	98.0	0.1688	98.5
λ_2^1	8.824	0.8335	90.6	0.1487	98.3	0.1161	98.7
Splitting and 2-prong substructure							
$z_g^\dagger$	2.041	2.041	0.0	0.2012	90.1	0.1972	90.3
$\tau_{21}^{(\beta=1)}$	10.50	6.577	37.4	0.5327	94.9	0.1412	98.7
Hadronization, multiplicity and fragmentation (IRC-unsafe)							
M	7.842	12.83	−63.7	0.2576	96.7	0.2621	96.7
n_{ch}	6.447	9.773	−51.6	0.1882	97.1	0.1872	97.1
f_{ch}	1.025	0.9435	7.9	0.1005	90.2	0.1121	89.1
p_T^D	8.342	4.029	51.7	0.2553	96.9	0.2841	96.6
q	2.027	0.9958	50.9	0.09290	95.4	0.1102	94.6

[†] IBU's purity binning produced a single bin, so IBU failed to unfold.

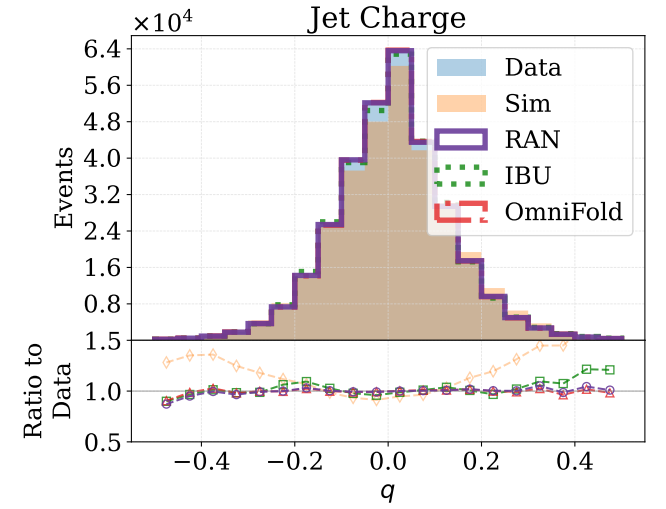
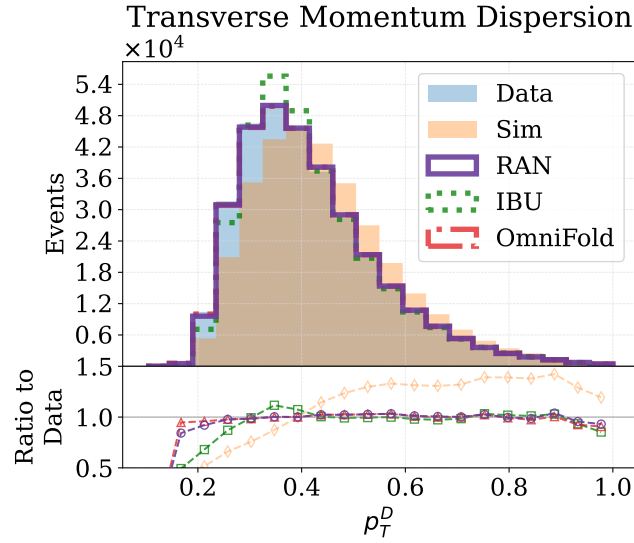
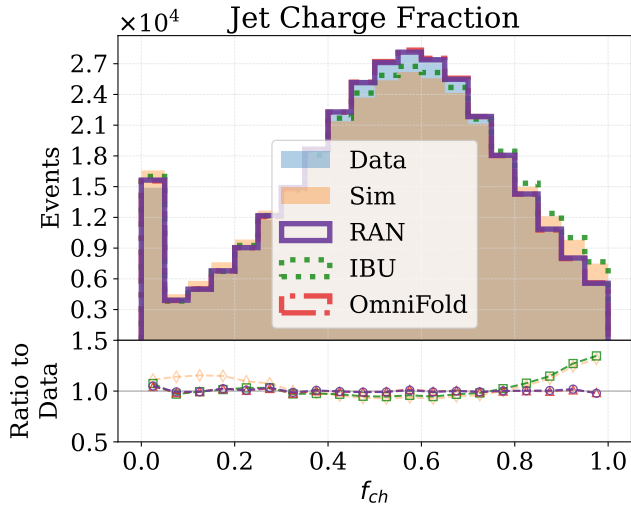
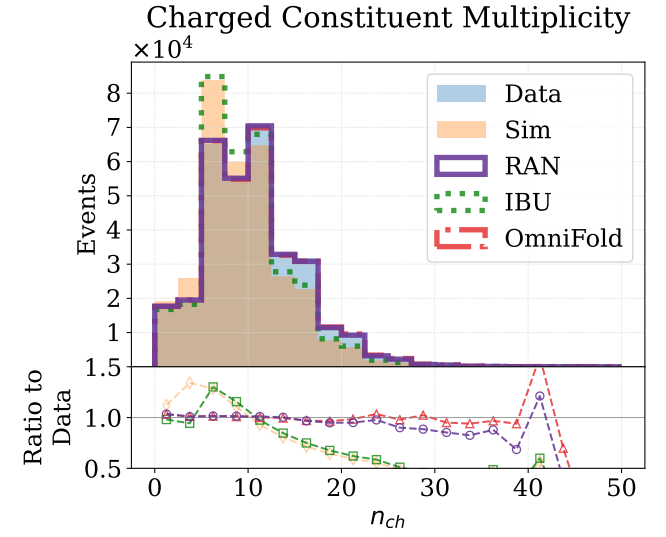
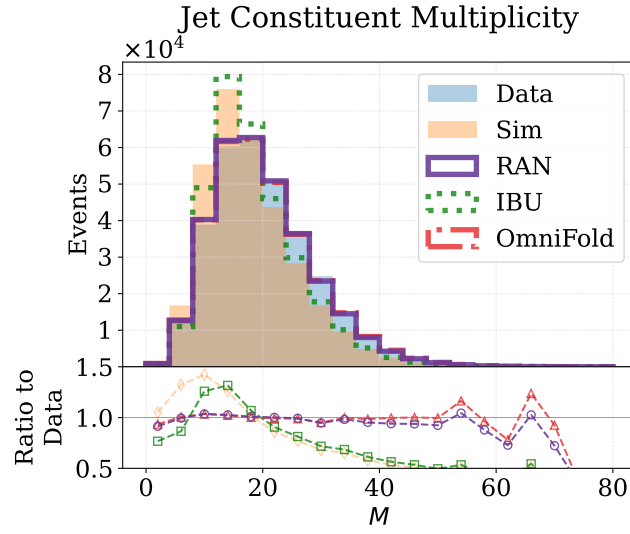
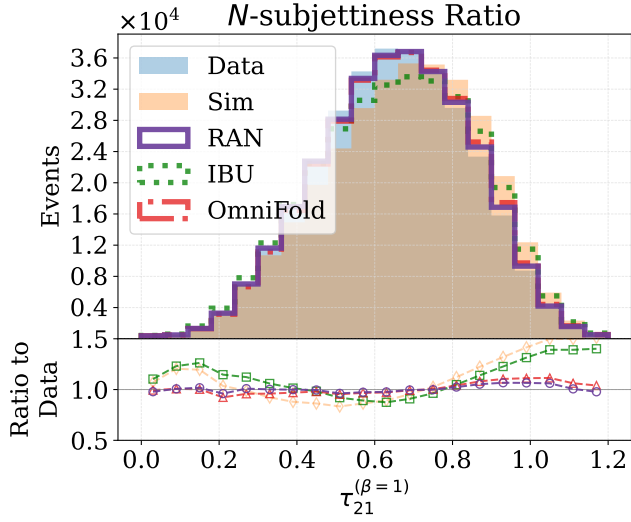
VLC divergence [$\times 10^3$]

[illegible]

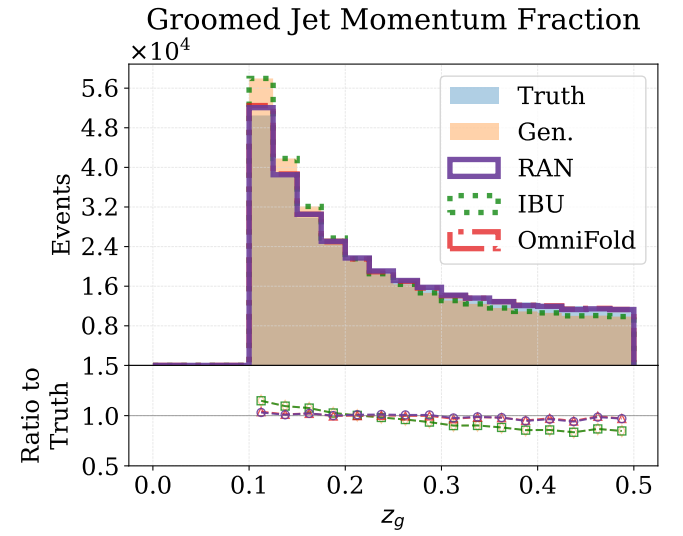
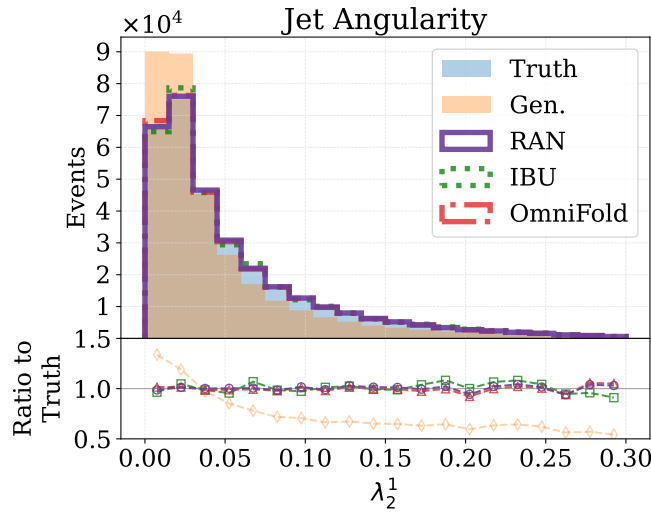
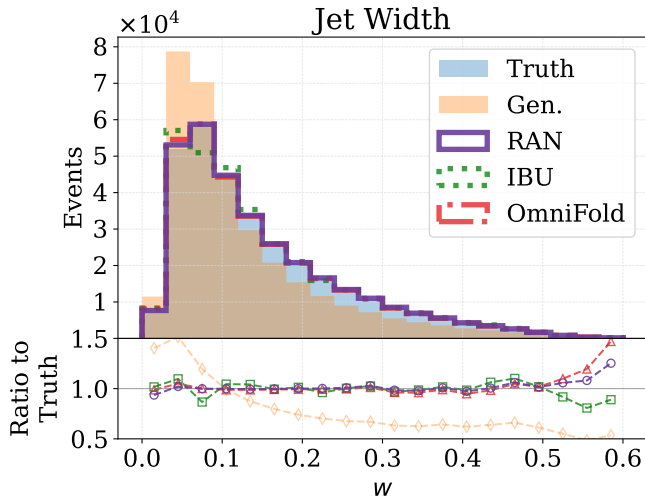
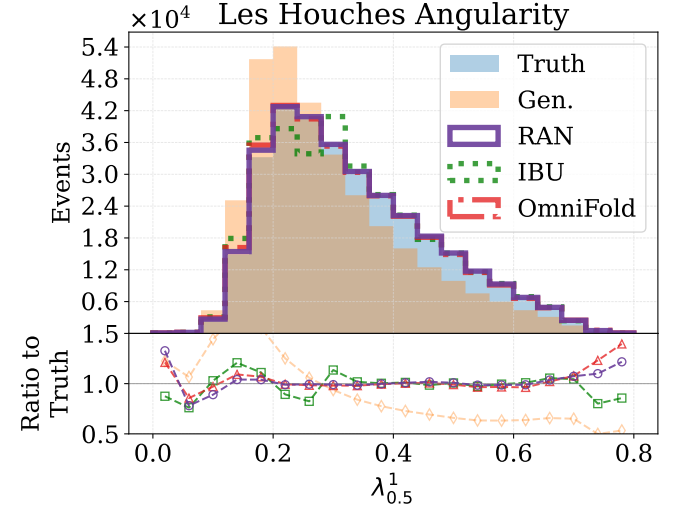
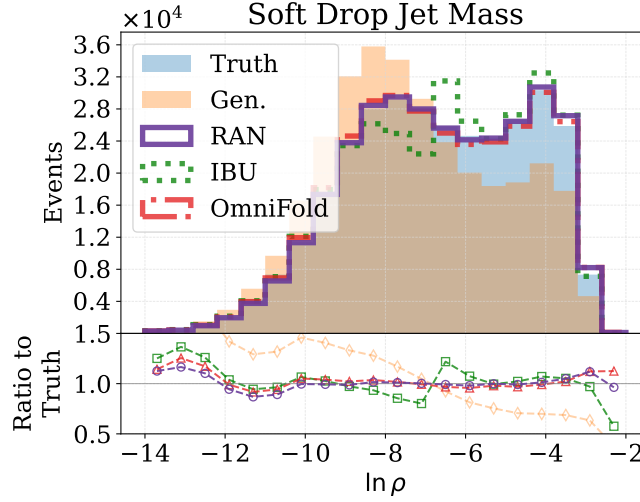
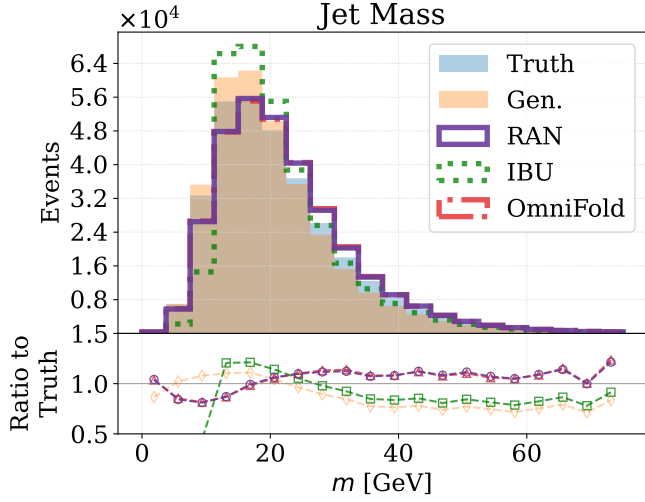
Detector Level (1 of 2)



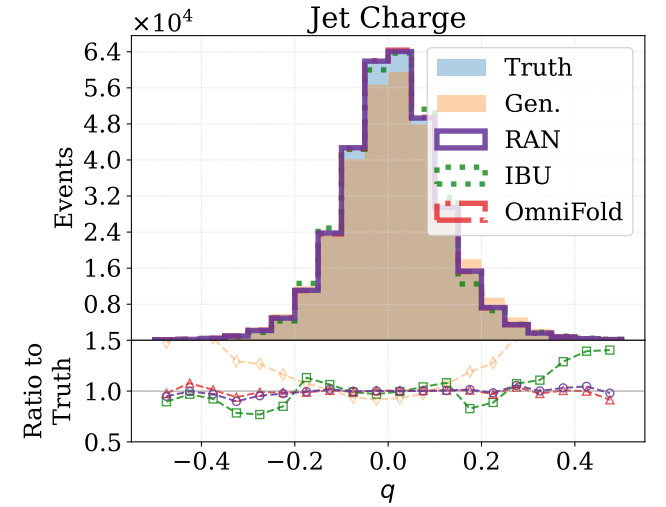
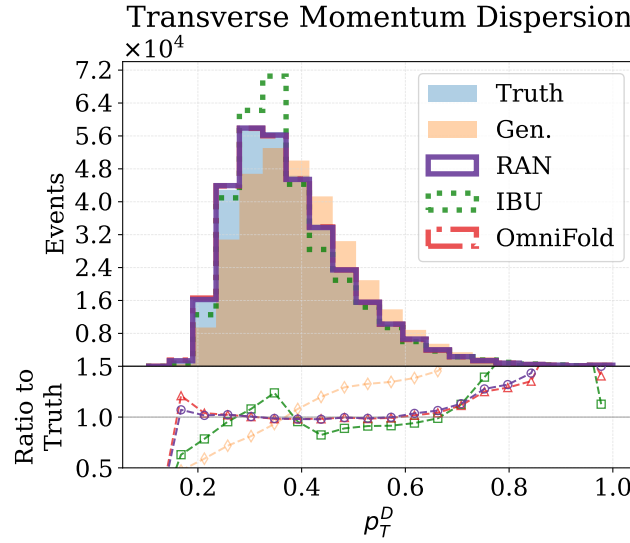
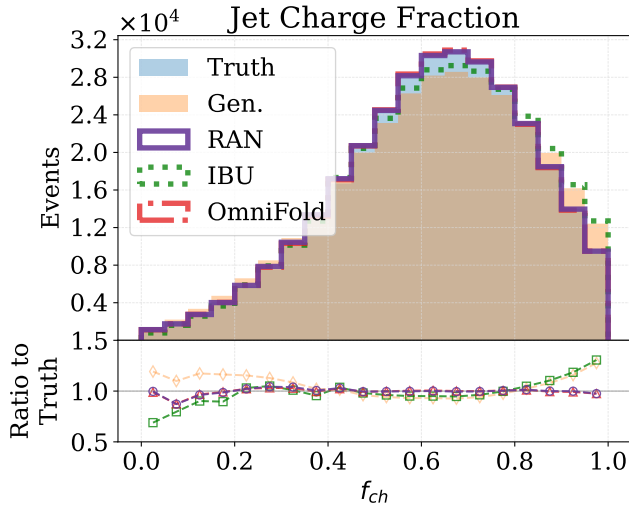
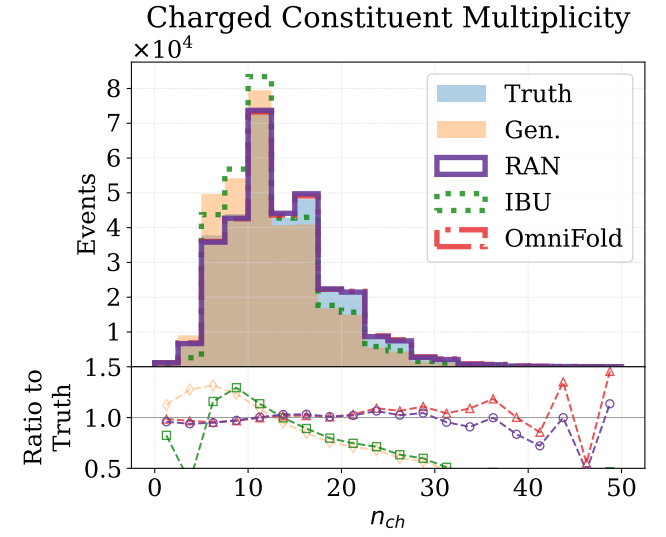
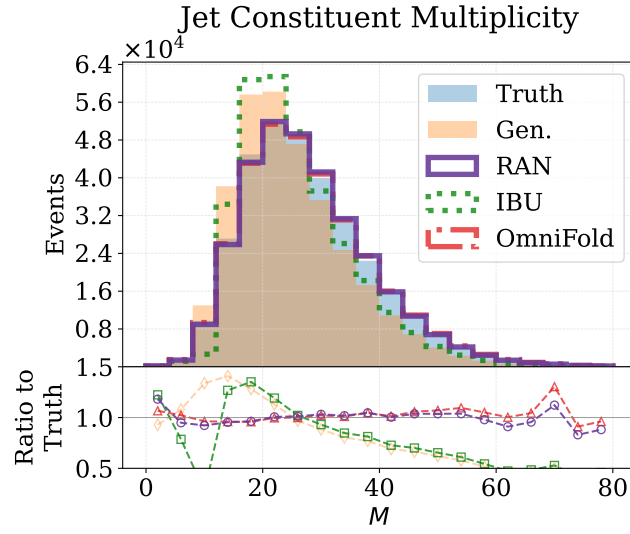
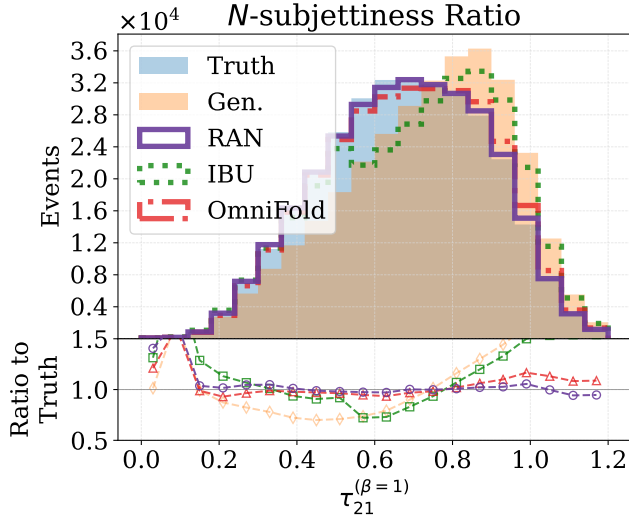
Detector Level (2 of 2)



Particle Level (1 of 2)

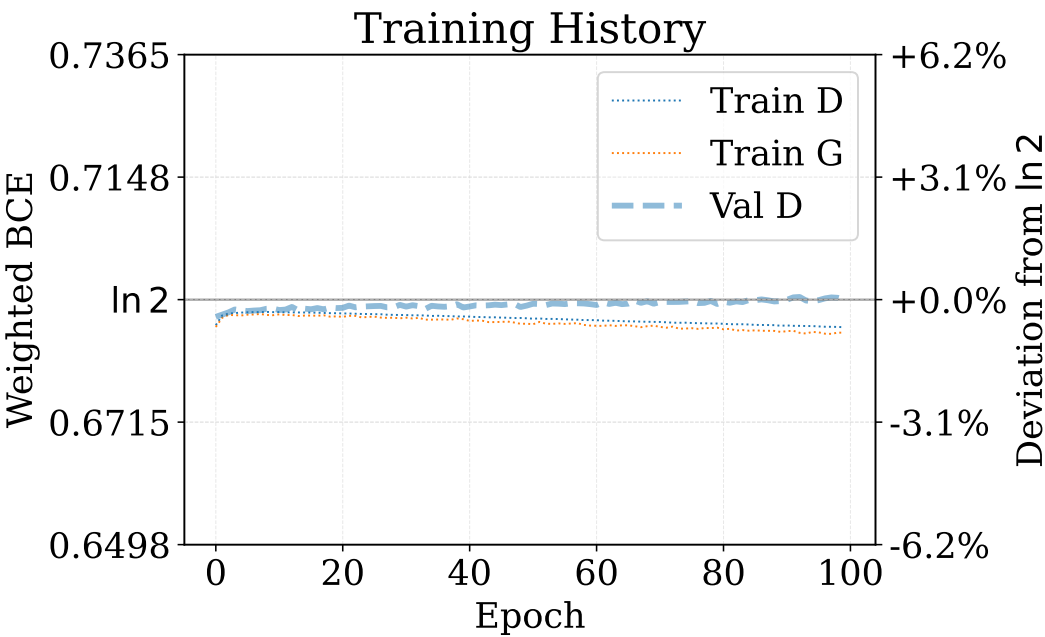


Particle Level (2 of 2)



Training diagnostics

Training losses



Selection criteria

